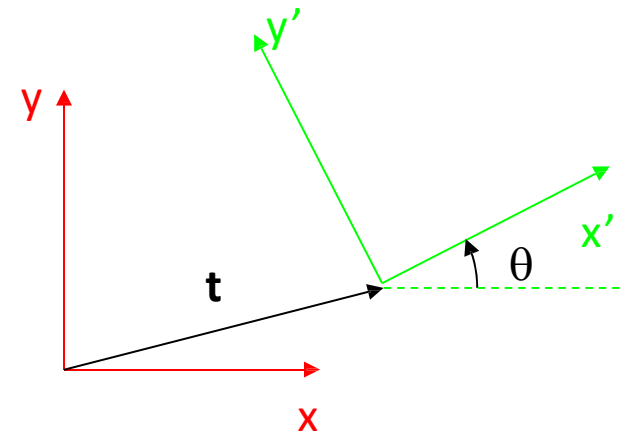


2D-2D Coordinate Transforms

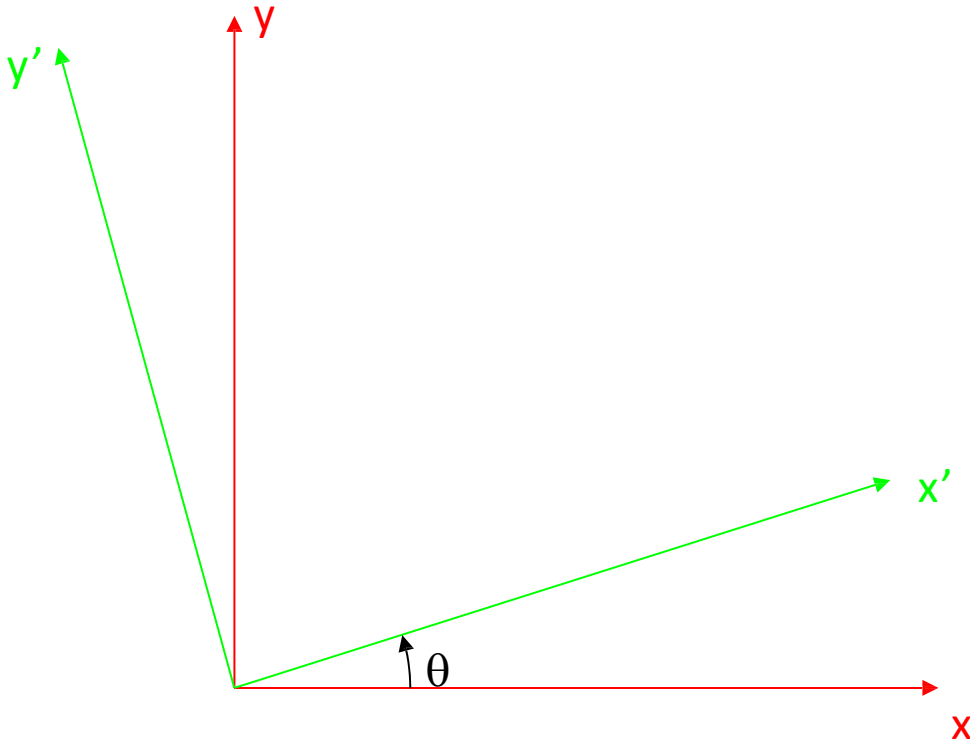
2D Rigid Frame Transformations

- The pose of one 2D frame with respect to another is described by
 - Translation vector $\mathbf{t}=(\Delta x, \Delta y)^T$
 - Rotation angle θ
 - Rotation can also be represented as a 2x2 matrix $\mathbf{R}$
- Object shape and size is preserved
- Number of degrees of freedom for a 2D rigid transformation?

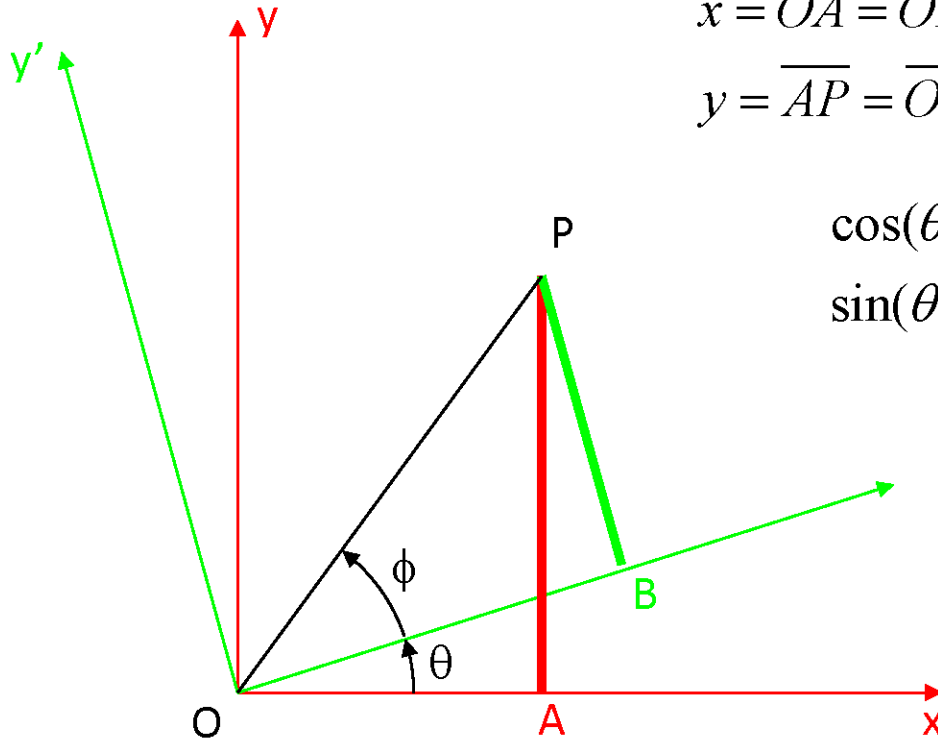


Rotations in 2D

- Let's derive the formula for a 2D rotation



Rotations in 2D



$$x = \overline{OA} = \overline{OP} \cos(\theta + \phi)$$

$$y = \overline{AP} = \overline{OP} \sin(\theta + \phi)$$

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$\sin(\theta + \phi) = \cos \theta \sin \phi + \sin \theta \cos \phi$$

$$x = \underbrace{\overline{OP} \cos \phi}_{x'} \cos \theta - \underbrace{\overline{OP} \sin \phi}_{y'} \sin \theta$$

$$= x' \cos \theta - y' \sin \theta$$

$$\text{Similarly, } y = x' \sin \theta + y' \cos \theta$$

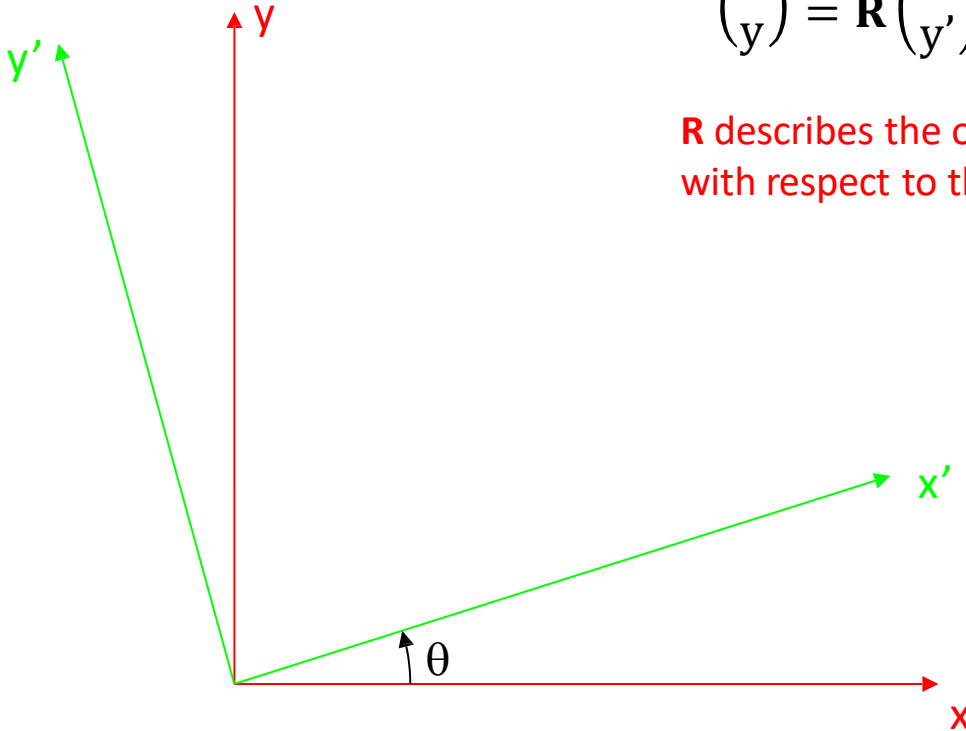
$$\text{So } \begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_{\mathbf{R}} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Rotations in 2D

$$\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{R} \begin{pmatrix} x' \\ y' \end{pmatrix} \quad \mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$\mathbf{R}$ describes the orientation of the “primed” frame (x', y') with respect to the “unprimed” frame (x, y)



- $\mathbf{R}$ is orthonormal
 - Rows, columns are orthogonal ($\mathbf{r}_1 \cdot \mathbf{r}_2 = 0$, $\mathbf{c}_1 \cdot \mathbf{c}_2 = 0$)
 - Transpose is the inverse; $\mathbf{R}\mathbf{R}^T = \mathbf{I}$
 - Determinant is $|\mathbf{R}| = 1$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \mathbf{R}^T \begin{pmatrix} x \\ y \end{pmatrix}$$

Homogeneous Coordinates

- Points can be represented using homogeneous coordinates
 - This simply means to append a 1 as an extra element
 - If the 3rd element becomes $\neq 1$, we divide through by it

$$\tilde{\mathbf{x}} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} sx \\ sy \\ s \end{pmatrix}$$

- Effectively, vectors that differ only by scale are considered to be equivalent
- This simplifies transform equations; instead of

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \mathbf{R} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \quad \mathbf{x}' = \mathbf{R}\mathbf{x} + \mathbf{t}$$

- we have

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad \mathbf{x}' = \mathbf{H}\mathbf{x}$$

$\mathbf{x} \in P^2$, where $P^2 = R^3 - (0,0,0)$ is called a 2D projective space

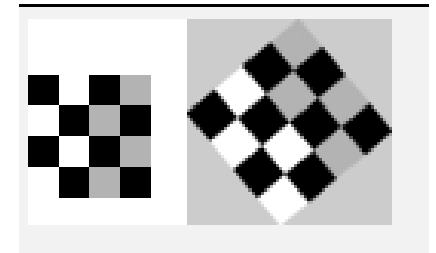
Example

- Transform the 2D point $x = (10, 20)^T$ using a rotation of 45 degrees and translation of (+40, -30).

Other 2D-2D Transforms

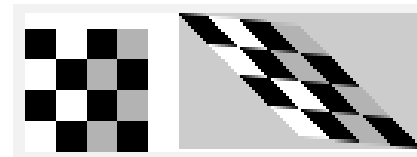
- Scaled (similarity) transform
 - preserves angles but not distances

$$\begin{pmatrix} x_B \\ y_B \\ 1 \end{pmatrix} = \begin{pmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix}$$



- Affine transform
 - Models rotation, translation, scaling, shearing, and reflection

$$\begin{pmatrix} x_B \\ y_B \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix}$$

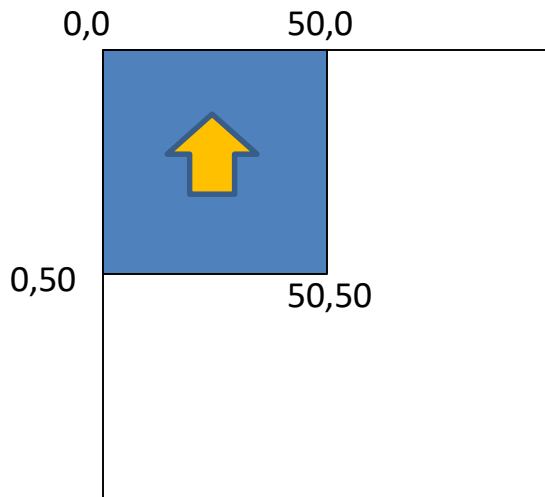


pairs of corresponding points needed?

Example

- Image “A” is modified by the affine transform below. Sketch image “B”

$$\begin{pmatrix} x_B \\ y_B \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0.25 & 1.5 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix}$$



A

Example Affine Warp



An example of placing an image inside another image using an affine transformation.

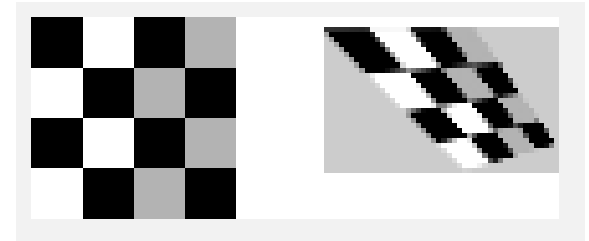
Projective Transform (Homography)

- Most general type of linear 2D-2D transform
- H is an arbitrary 3x3 matrix

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad \tilde{\mathbf{x}}' = \mathbf{H} \tilde{\mathbf{x}}$$

- We still need to divide by the 3rd element

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} x_1 / x_3 \\ x_2 / x_3 \\ 1 \end{pmatrix}$$



As we will see later, a homography maps points from the projection of one plane to the projection of another plane

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix}_{2 \times 3}$	2	orientation	
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}_{2 \times 3}$	3	lengths	
similarity	$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \end{bmatrix}_{2 \times 3}$	4	angles	
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}_{2 \times 3}$	6	parallelism	
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}_{3 \times 3}$	8	straight lines	

Table 2.1 Hierarchy of 2D coordinate transformations. Each transformation also preserves the properties listed in the rows below it, i.e., similarity preserves not only angles but also parallelism and straight lines. The 2×3 matrices are extended with a third $[0^T \ 1]$ row to form a full 3×3 matrix for homogeneous coordinate transformations.